

Computational Methods for Single-Cell RNA-Seq Analysis of *Arabidopsis thaliana*

Part 1 — Computational Environment and Quick Start

SingleCell Pipeline

2026-06-24

Materials and Methods

Materials

The pipeline source code is available at <https://github.com/cliford2001/SingleCell>:

```
— bash —
git clone https://github.com/cliford2001/SingleCell.git
cd SingleCell
```

The repository is meant to sit inside a working directory that also holds the raw data and the pipeline's output folder — for instance `~/your-workdir/SingleCell`, with `~/your-workdir/resultados` alongside it. That parent directory, not the repository alone, is what gets mounted into the container, so that code, input matrices, and results are all visible together at `/workspace`.

Docker packages R, Python, and every pipeline dependency into one portable **image** (`matigara/scrnaseq:latest`), so the analysis runs identically on any machine with Docker installed. Two equivalent ways to start it:

Option 1 — single command

```
— bash —
docker run -it --rm \
  -v ~/your-workdir:/workspace \
  matigara/scrnaseq:latest /bin/bash
```

Pulls the image if missing and opens a shell. Replace `~/your-workdir` with the actual path to that working directory if it lives somewhere else.

Option 2 — the repository's docker-compose.yml

Run as-is from the repository root:

```
bash  
docker compose pull  
docker compose run --rm r bash
```

`pull` only reads that file and downloads the image it names — unrelated to `git`, which just placed the file on disk. Same `~/your-workdir` → `/workspace` mount as Option 1.

Either way, `/workspace` **is** `~/your-workdir` on the host: files written there persist after the container exits.

Once inside the shell, Chapter 1-2 commands are typed into `R`, and Chapter 3 commands into `Python`:

```
R  
R
```

```
python  
python3
```

Methods

Chapter 1 — Single-cell processing

Running Option 1 opens a self-contained terminal inside the container, separate from the host shell it was launched from:

```
— bash —
docker run -it --rm \
  -v ~/your-workdir:/workspace \
  matigara/scrnaseq:latest /bin/bash
```

R is then started from that terminal, on its own:

```
— R —
R
```

Chapter 1 (`capitulo1_single_cell.R`) runs entirely from this session.

Initialization

Paths are set once at the top of the script: `PIPELINE_DIR` points to the helper scripts, and `DATA_DIR/base_dir` define where input data are read from and where every result file is written. Three helper scripts are then sourced — package loading, custom Seurat plotting utilities, and the core analysis functions used throughout the chapter — before fixing the random seed (1807) for reproducibility, disabling `Seurat.allow.s4` for Seurat 5 compatibility, and setting the working directory to `DATA_DIR`.

```
— R —
PIPELINE_DIR <- "/workspace/ScRNASeq-Docker/workflow"
DATA_DIR      <- "/workspace/."
base_dir      <- file.path(DATA_DIR, "resultados")

source(file.path(PIPELINE_DIR, "load_libraries.R"))
source(file.path(PIPELINE_DIR, "custom_seurat.R"))
source(file.path(PIPELINE_DIR, "ScRNA_Analysis_Functions.R"))

set.seed(1807)
options(Seurat.allow.s4 = FALSE)
setwd(DATA_DIR)
```

Sample definition and data loading

Three samples spanning a nitrogen-availability gradient were analyzed — `Sample_0N`, `Sample_05N`, `Sample_5N` (conditions 0N, 0.5N, 5N) — each loaded from its CellRanger

